

正则表达式 Regular Expressions

- 正则表达式是另一种**NFA相关**的语言定义表示法，可以将其视为NFA表示法“**用户友好**”的替代方案。Regular expressions are another type of language-defining notation related to nondeterministic finite automaton and can be thought of as a "user-friendly" alternative to the NFA notation.
- 正则表达式可以定义正则语言并提供一种声明性的方式来表达系统可以接受的字符串。因此，正则表达式经常作为处理字符串的许多系统的输入语言。Regular expressions can define the regular languages and offer a declarative way to express the strings the systems can accept. Thus, regular expressions always serve as the input language for many systems that process strings.

正则表达式在实际应用中的例子

- 例如，文本搜索工具grep(Global Regular Expression and Print):

```
grep -Eo '[0-9]{17}[0-9Xx]' file.txt
```

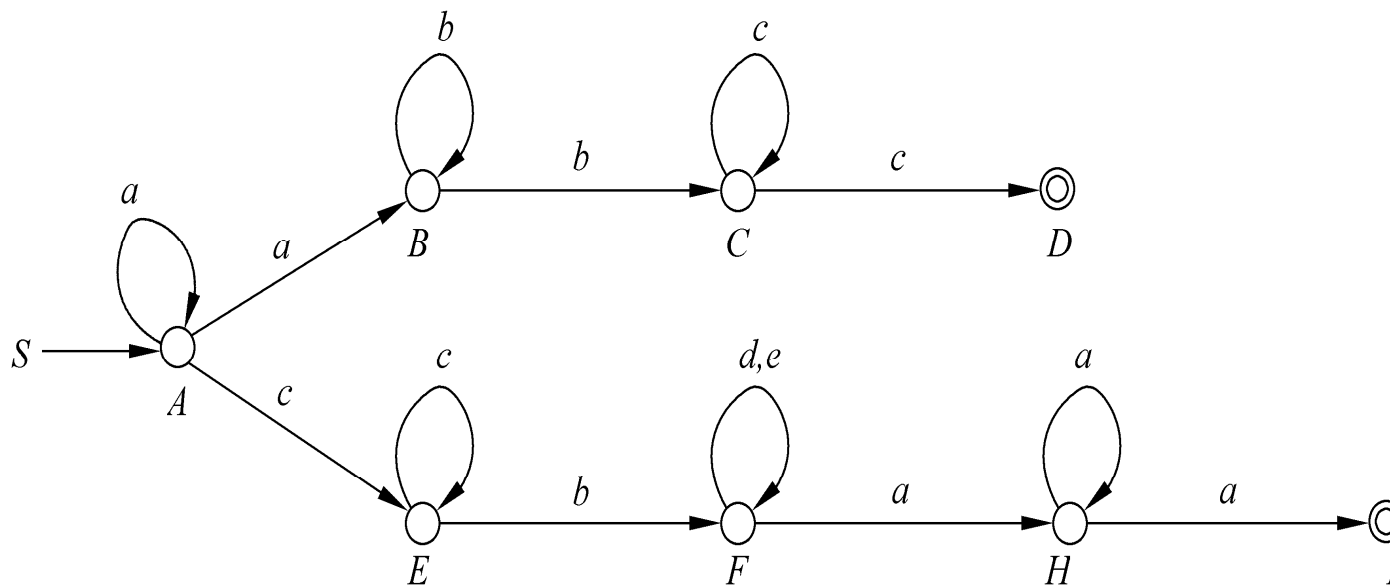
- `[0-9]{17}[0-9Xx]` 是一个正则表达式，它的意思是匹配包含 18 个字符的数字或字母序列，其中前 17 个是数字，最后一个可以是数字、大写字母 X、小写字母 x 中的任意一个。

正则表达式——简洁性

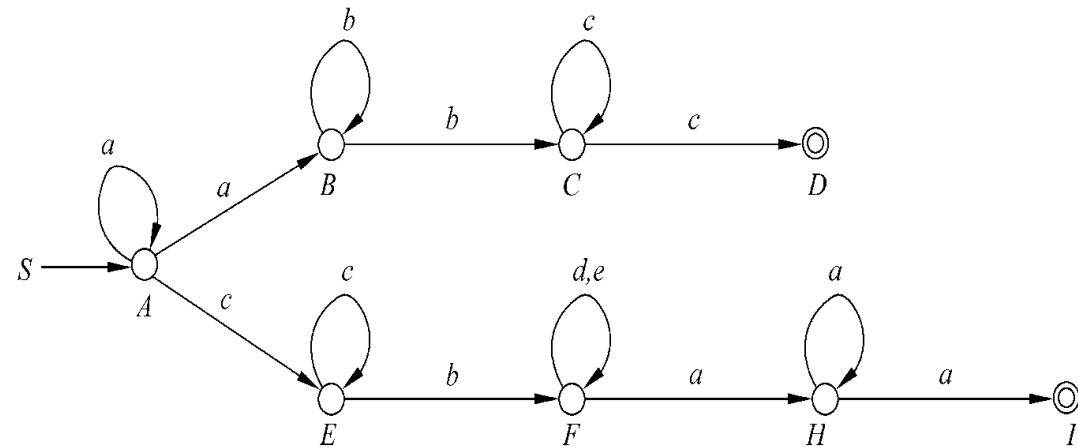
考虑语言 $\{a^n b^m c^k \mid n, m, k \geq 1\} \cup$

$\{a^i c^n b x a^m \mid i \geq 0, n \geq 1, m \geq 2, x \text{ 为 } d \text{ 和 } e \text{ 组成的串}\}$

- 接受此语言的NFA M



正则表达式——简洁性



- 计算该NFA中各状态的 $\text{set}(q)$ 集:

$$\text{set}(A) = \{a^n | n \geq 0\} = \{a\}^*$$

$$\text{set}(B) = \text{set}(A)\{a\}\{b^n | n \geq 0\} = \{a^n a b^m | m, n \geq 0\} = \{a\}^* \{a\} \{b\}^* = \{a\}^+ \{b\}^*$$

$$\text{set}(C) = \text{set}(B)\{b\}\{c\}^* = \{a\}^* \{a\} \{b\}^* \{b\} \{c\}^* = \{a\}^+ \{b\}^+ \{c\}^*$$

$$\text{set}(D) = \text{set}(C)\{c\} = \{a\}^+ \{b\}^+ \{c\}^* \{c\} = \{a\}^+ \{b\}^+ \{c\}^+$$

$$\text{set}(E) = \text{set}(A)\{c\}\{c\}^* = \{a\}^* \{c\} \{c\}^* = \{a\}^* \{c\}^+$$

$$\text{set}(F) = \text{set}(E)\{b\}\{d,e\}^* = \{a\}^* \{c\}^+ \{b\} \{d,e\}^*$$

$$\text{set}(H) = \text{set}(F)\{a\}\{a\}^* = \{a\}^* \{c\}^+ \{b\} \{d,e\}^* \{a\} \{a\}^* = \{a\}^* \{c\}^+ \{b\} \{d,e\}^* \{a\}^+$$

$$\text{set}(I) = \text{set}(H)\{a\} = \{a\}^* \{c\}^+ \{b\} \{d,e\}^* \{a\}^+ \{a\}$$

正则表达式——简洁性

$$\begin{aligned} L(M) &= \text{set}(D) \cup \text{set}(I) \\ &= \{a\}^+ \{b\}^+ \{c\}^+ \cup \{a\}^* \{c\}^+ \{d, e\}^* \{a\}^+ \{a\} \end{aligned}$$

根据集合运算的定义, $\{d, e\} = \{d\} \cup \{e\}$ 。

从而, $\{d, e\}^* = (\{d\} \cup \{e\})^*$ 。

这样可以将 $L(M)$ 写成如下形式:

$$L(M) = \{a\}^+ \{b\}^+ \{c\}^+ \cup \{a\}^* \{c\}^+ (\{d\} \cup \{e\})^* \{a\}^+ \{a\}$$

进一步省略 $\{\}$, $L(M)$ 可记作:

$$L(M) = a^+ b^+ c^+ + a^* c^+ (d+e)^* a^+ a = aa^* bb^* cc^* + a^* cc^* (d+e)^* aaa^*$$